# NETWORK & SECURITY

## CRYPTOGRAPHY

DATE: 25/01/2026

### What is Crypto?

It is both art and science of secret communication

Design of secret communication system

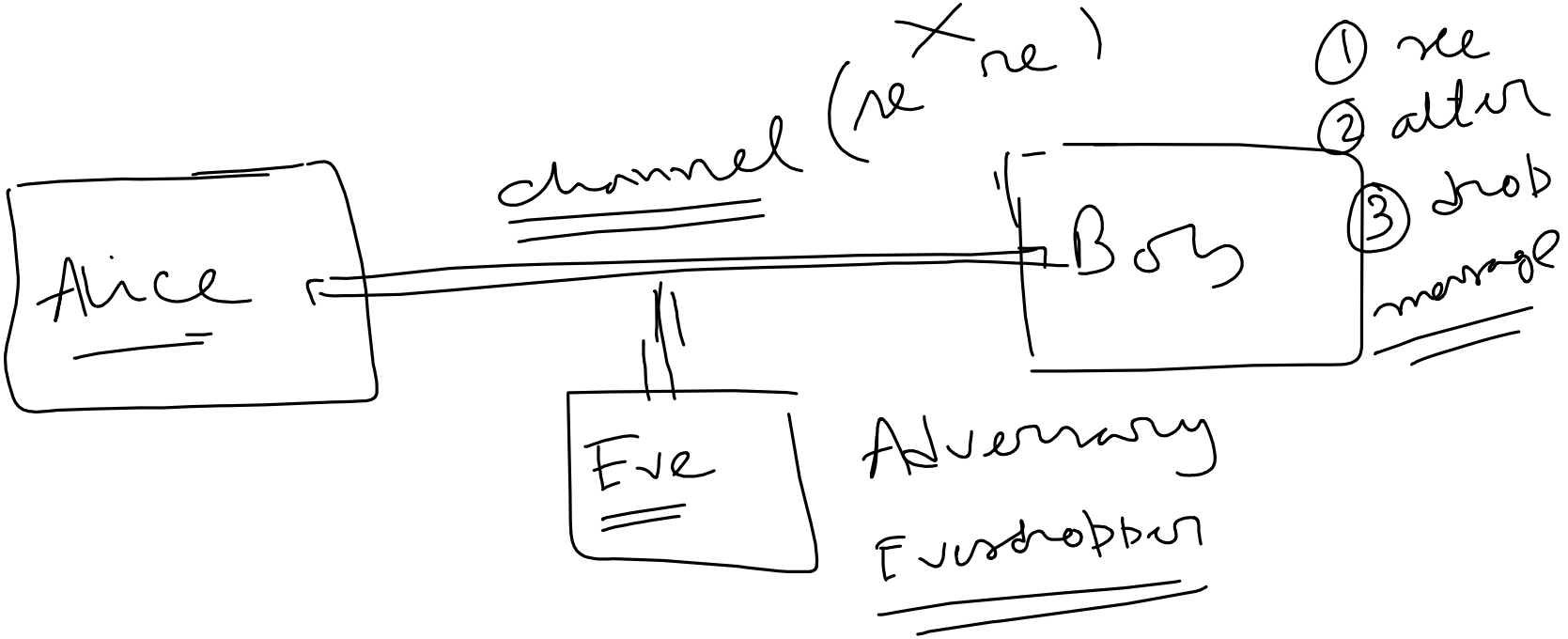
- Rationale behind the art

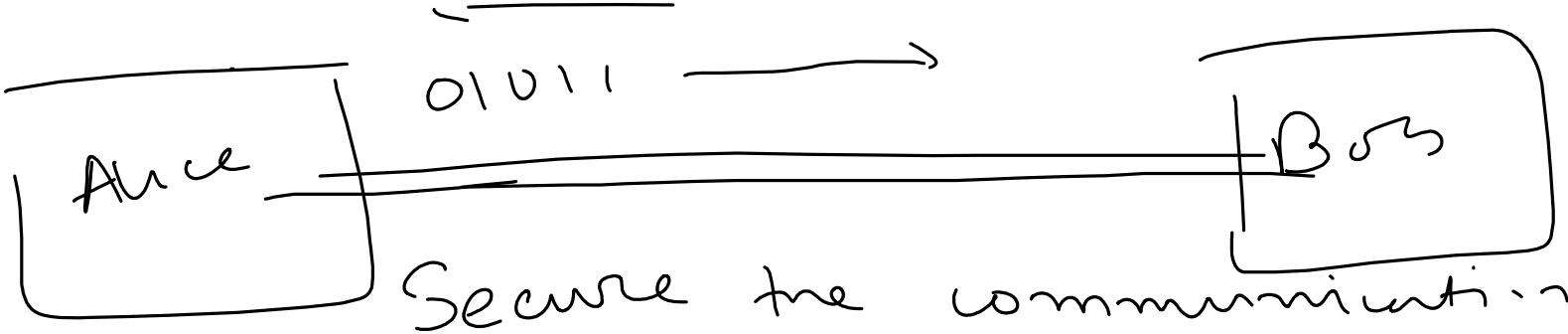
CRYPTOLOGY

- CRYPTOGRAPHY — DESIGN
- CRYPTO-ANALYSIS — HOW TO BREAK

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Now






Secure the communication.

## Goals of cryptography

C — Confidentiality (Encryption)

I — Integrity ($\overset{A}{m}, \overset{B}{m}$)

A — Authentication — Some party sends message m, we must get the authenticated message.

N — Non-repudiation | commitment

Sign a check. Later I can't say that I did not sign my signature in my commitment.

# Definition of Encryption

It has 3 algorithms

1. Gen — This also generates key randomly.
2. Encryption (Enc): Given a key and a message (m) plaintext. outputs a ciphertext (c)
3. Decryption (Dec): Given a key(k) and a ciphertext (c) outputs a message (m)

Soundness of this def: $Dec(K, Enc(K, m)) = m$

Definition

$M$ : Message Space or plaintext space.

$K$ : Key Space

$C$ : Ciphertext Space

Now There are two types of Encryptions

- Symmetric
  - Historic (Same key)
- Asymmetric
  - Newer concept (Different keys are created among users)

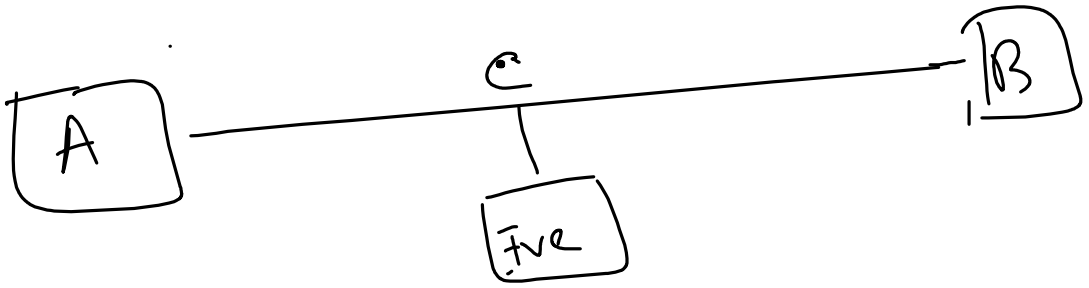
# Symmetric Encription

$c = Enc(K, m)$ ;

$m = Dec(K, c)$ ;

Alice and Bob both have same copy of the key

Eve is the adversary who is trying to know what the key is. or what the message m is



# Historical Examples of Cryptography

## Caesar Cipher

A — D
B — E
C — F
⋮
Z — C

$Enc(m) = (m+3) \bmod 26$

$m \in \{A, B, C, D, \ldots Z\}$ 26 letters

$Enc(A) = D \bmod 26$

Mod operation

$= (a+b) \bmod 26$

ex: $(10+5) \bmod 26 = 15$ $\quad a = 10 \quad b = 5$

$15 < 26$

$a = 20, b = 10$

$\Rightarrow (20+10) \bmod 26$

$= 30 \bmod 26 = 4 \bmod 26$

A (1) → D (4)

B (2) → E (5)

Z (26) → C (26+3) mod 26

= (3)

Example

BOOK

ERROR

Receiver

Receiver

ERRN

shift 3 letters back

to BOOK

DEC (C) = (C − 3) mod 26

It is recurse or not? Why not recurse

cipher text

and find any meaningful word out of this.

we need to check the shift

ERRN = CPPL 2places | BOOK 3places

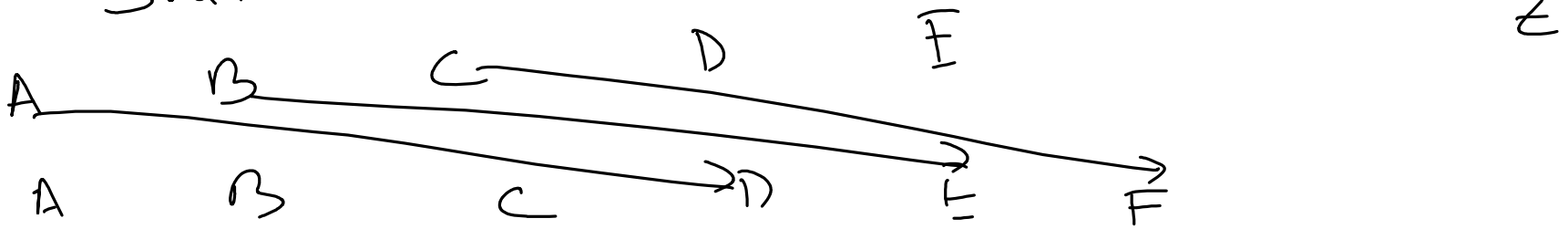
# SHIFT CIPHER

$Enc(K, m) = (m + 12) \bmod 26$

$m \in \{A, B, C, \ldots, Z\}$

$K \in \{1, \ldots, 26\}$

Caesar Cipher is one ~~instance~~ instantiation of Shift Cipher



# How do we break Shift Cipher?

now

$$Dec(k, c) = (c - k) \bmod 26$$

## Exhaustive Search

c , K=1 , K=2 , K=3 , ... K=26

check whether we get anything meaningful.

26

in between we will get some meaningful sentence

K, one of them makes a

c

So, What is the message space?

$m \in \{A, B, \ldots, Z\}$ 26 letters.

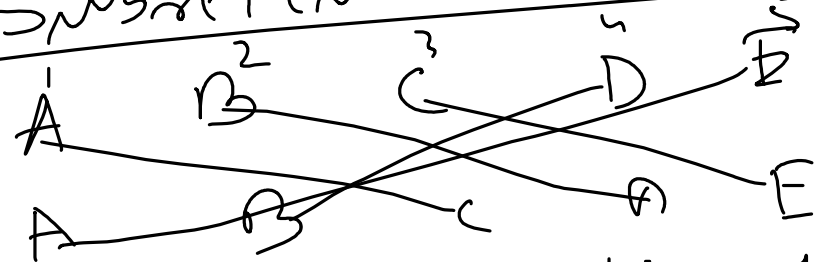
Key space $K \in \{1, 2, \ldots 26\}$ 26

26 is a very small number

C-1< 26 times english alphabet

Substitution Cipher

| | |
|---|---|
| 1 | 3 |
| 2 | 4 |
| 3 | 5 |
| 4 | 2 |
| 5 | 1 |




M= (5)

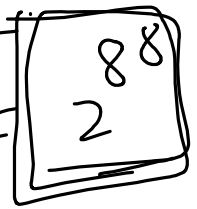
permutations we can think of 15

26 letter $26! \approx 2^{88}$

C =

C — try

try all permutation
9 will take @26! time




very big number

10 2 years

exhaustive search

we will end up doing this for many years.

English alphabet has some statistical data.

, where e → comes more frequently

a, b, c, d, e, ... z has statistical frequency

## Breaking Strategy

From our ciphertext find the frequency of letters and match the frequency with the known frequency of the english alphabet.

c = p. has highest fre
p → i/e which ever has most fre.
l → s

This way we can find the mapping of letters of plaintext to the letters of ciphertext.

And this doesn't take much time

Why?

a, b, c, d, ...

$p_1, p_2, p_3, \ldots p_{26}$ the frequency of the ciphertext we have

$d_1, d_2, \ldots d_{26}$

$$\sum_{i=1}^{26} p_i^2 \simeq 0.0655$$

$$\sum_{i=1}^{26} p_i d_j \simeq 0.0655$$

all the 'K' (1, 26)

$j = i + K$ This is for encryption

We all have mobiles

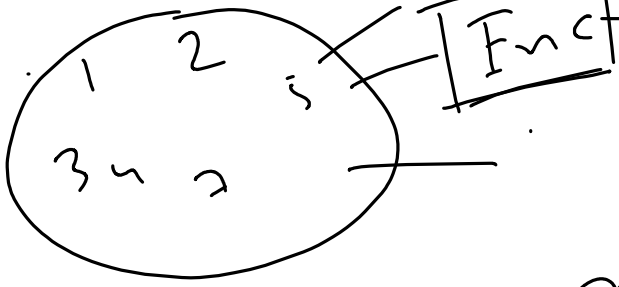
- cryptography
  - SIM has some key by the company — JIO, VODAFONE
  - Insert the SIM to the mobile
  - It connects and validates it-ly identity with the nearest mobile tower

Bluetooth — pair a bluetooth device to the mobile we check the manage to be connect.

# Kerckhoff's principle :-

| Enc, | Dec, | Gen, | m, | c, | K, | 'r' | it parameters |
|---|---|---|---|---|---|---|---|

What are the things should be kept secret and what are kept public ?




algorithm is not safe anymore

→ Everything should be kept public except Key (K) and the message (m) → c

1. Caesar's cipher
2. Shift Cipher
3. Substitution / Permutation Cipher
4. Affine Cipher

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$$Enc(m) = (am + b) \bmod 26$$

$m \in$ alphabets english

$$Dec(c) = \left(\frac{c-b}{a}\right) \bmod 26$$

$a = 10$
$m = D = 4$
$b = 5$

$c = Enc(4) = (10 \times 4 + 5) \bmod 26$
$= (40 + 5) \bmod 26$
$= 19 \bmod 26$

$$Dec(C) = \frac{C-b}{a} \mod 26$$

$$= \frac{19-5}{10} \mod 26$$

$m = 4$

$$m' = 14/10 \mod 26$$

$m \neq m'$ This is not correct.

We need $g.c.d(a, 26) = 1$ or a must be co-prime to 26. So we will discuss this in details when we study GCD in next classes.

# H.U Gpwn

$$\begin{array}{c|cc|cc|cc|c} x_1 & x_2 & & x_m & x_{m+1} & \cdots & x_{2m} & x_{2m+1} & \cdots & x_{3m} \\ \hline y_1 & y_2 & & y_m & y_{m+1} & & y_{2m} & y_{2m+1} & & y_m \end{array}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \\ y_m \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & & a_{2m} \\ & & & \\ a_{m1} & & \cdots & a_{mm} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \\ x_m \end{bmatrix}$$

$$Y = AX$$

$$m \times 1 \quad m \times m \quad m \times 1$$

$$y_1 = (a_{11} x_1 + a_{12} x_2 + \cdots + a_{1m} x_m)$$

$$\begin{bmatrix} x_1 \\ x_2 \\ \\ x_m \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} & & b_{1m} \\ & & & \\ b_{m1} & b_{m2} & & b_{mm} \end{bmatrix} \begin{bmatrix} y_1 \\ \\ y_m \end{bmatrix}$$

$$Y = A \cdot X \quad , \quad X = B \cdot Y$$

$\Rightarrow$ A and B are inverse to each other

mxm modulus $(n)$. $A \cdot B = I_m \bmod n$

So, A, B are invertible matrices

$m \times m = m^2$ multiplications are needed

# Modes of Attack

1. Ciphertext Only
2. Known plaintext (Hill cipher) } Very practical
3. chosen plaintext
4. chosen Ciphertext } Strict

Find none ciphertext and ask alice on bob to decrypt

Adversary can impersonate Bob and ask Alice to decrypt.

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

What is $A$?

# Vigenere Cipher (OTP)

$M$ $\{0,1\}^{\ell}$ $\ell$ Length bit stream of 0, 2

$K$ $\{0,1\}^{\ell}$ $r$ $r$ $r

XOR operation

| x \ y | 0 | 1 |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 1 | 0 |

$$
\begin{array}{r}
1\ 1\ 0\ 1 \\
+\ 1\ 1\ 1\ 0 \\
\hline
2\ 2\ 1\ 1 \\
\hline
0\ 0\ 1\ 1
\end{array}
\qquad \text{mod } 2 \qquad \text{mod } 2
$$

$x \bmod 2$

$5 \bmod 2 = 1$

$$
\begin{array}{r|l}
2 & 5 \\
 & 4 \\
\hline
 & 1
\end{array}
\quad (2
$$

$$Dec(c, k) = \overset{m}{\circledcirc} c \oplus k = m$$

$$= 0011 \oplus 1101$$

$$= 1110 = m$$

Symmetric key

(1) Alice and Bob share a key in advance

A Alice wants to send $m$, $\Rightarrow$ Alice

$Enc(k, m) = k \oplus m = c$ to Bob

Ans Bob decrypts $c$ to $m$ by $c \oplus k$ gets back $= m$

Size of key is length 'l'

Size of m = length 'l'

So, to send l bit message we need to send l bit keys.

If we want 1 GB data to send we need 1 GB in keys

This is not a practical scheme. (1977)

Can we use a key multiple times?

$c_1 = m_1 \oplus k$

$c_2 = m_2 \oplus k$

$\Rightarrow c_3 = c_1 \oplus c_2 = \boxed{m_1 \oplus m_2}$ addition modulo 2.

## Perfect Secrecy

First of all OTP is perfectly secure

$M \in \{0,1\}^{\ell}$ where

$|M| = 2^{\ell}$

$= |C| = 2^{\ell}$

$|K| = 2$

$m_1 = p_1$

$m_2 = p_L$

$m = b$

$\forall$ all probability distributions over $M$, and for

~~Plaintext~~ $m$

$P_\pi[M=m \mid C=c] = P_\pi[M=m]$

# Perfect Secrecy

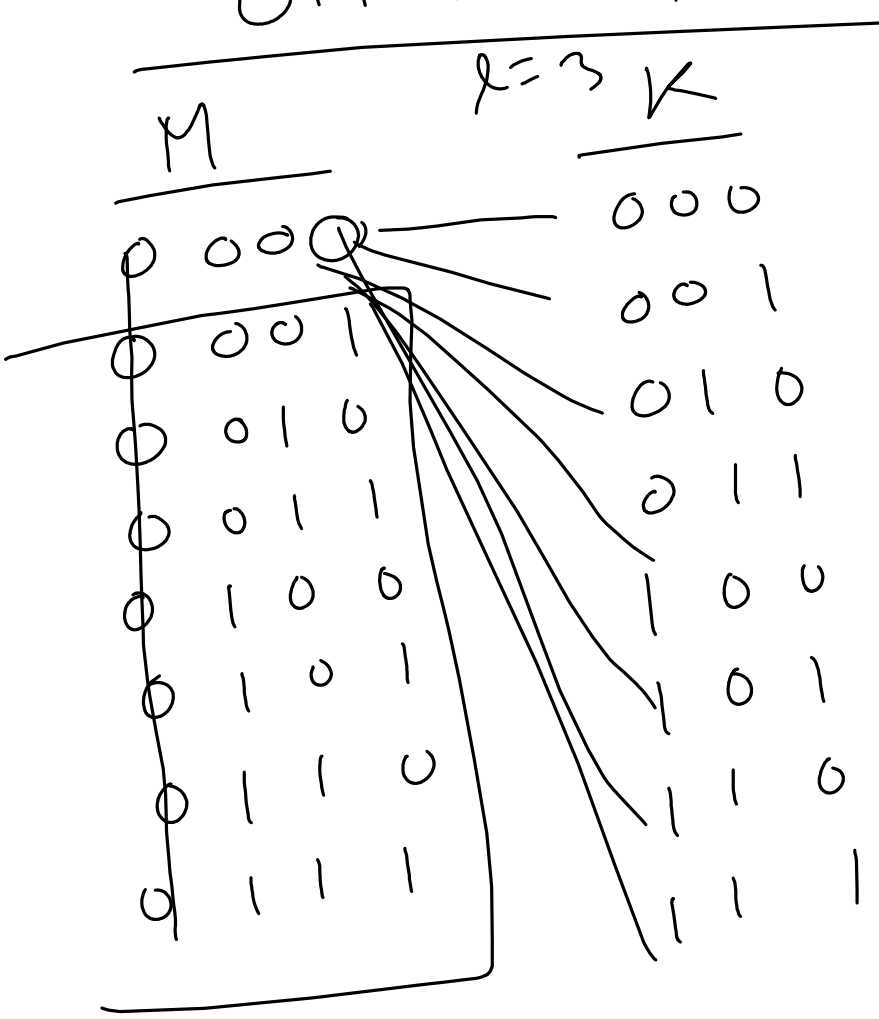
$$Pr[M = m \mid C = c] = Pr[M = m]$$

Here $M$ is the random variable denoting the plaintext

Similarly for Ciphertext, $C$.

→ The distribution of $M$ and $C$ are mutually independent.

# OTP is perfectly secure crypto scheme




C

000
001
010
011
101
101
110
111

C

110

The probability that plaintext is

001

$\Rightarrow K =$ 111

Keys are drawn uniformly at random. So, probability is equal to $1/2^3 = 1/8$

When we say uniformly at random dist.

$\Rightarrow$ The samples are drawn uniformly from

if there are 5 sample points

$\Rightarrow$ probability of each sample point = $1/5$

Random = Unpredictable.

We know that any of 5 samplepoints can be drawn but which one we don't know.